

Case Study Rubric

DS 4002 – Spring 2026 – Stephanie Diau

Due: TBD

Submission format: Upload link to GitHub repository in UVA Canvas

Individual Assignment

Why am I doing this? This case study is a great opportunity to apply data science and statistical techniques on real-world time series data. As you work through this case study, you will be able to explore different methods and develop a deeper understanding of time series analysis.

What am I going to do? You will explore if song features, specifically those released after the year 2000, display monthly seasonal patterns using a Spotify dataset. You will use materials (data, scripts, etc.) from the following GitHub repository: <https://github.com/stephaniediau/DS4002-CS3>. This will serve as your “source repo”. In this case study, you will determine whether there is significant evidence of seasonal patterns based on the month of the year a song was released, specifically focusing on the following features: Acousticness, Speechiness, Valence, Tempo, and Duration. In summary, your task is to essentially produce results using materials found in the source repo, but with a narrower time frame (2000-2020 instead of 1980-2020) and discuss your findings in a written report.

Deliverables include:

- A Github repository that contains necessary code and data
- A PDF of a written report detailing your findings (included in repository)

This will be submitted electronically via a link in Canvas.

Tips for success:

- In your repository, name your files descriptively
- Your written report does not have to be extremely professional in writing quality, but it should show considerable effort and thoughtful thinking

How will I know I have succeeded? You will meet expectations on this case study when you successfully follow the criteria in the rubric below.

Formatting	• Submit the following on Canvas: ◦ Your GitHub repository, which should contain the following: ▪ A README.md file ▪ A LICENSE.md file (default to MIT) ▪ A DATA folder ▪ A CODE folder ▪ An OUTPUT folder ▪ Your written report (PDF)
Written Report	• This document should be at least 2 pages in length, use a legible font/font size, and contain: ◦ An introduction paragraph:

	■ Discuss the topic and why it might be important or interesting to explore and analyze ■ Mention the research question that you explored ○ An analysis paragraph: ■ Introduce and describe the SARIMA time series model and how it is relevant to this analysis ■ Describe the analysis process and any created code ○ A results paragraph: ■ Present your results and interpret in context using model metrics (AIC/BIC, p-values, etc) ■ State your final conclusion ○ A reflection paragraph: ■ Reflect on your experience completing this case study; document any challenges and/or successes ■ Discuss anything you learned or would do differently next time in a similar project or case study ○ A list of references on a new page (does not count towards page minimum) ■ Use IEEE Documentation type ■ Include the source repo as well
README.md	● Include a brief (3-4 sentences) description of the case study; refrain from discussing results as that will be included in your final written document
DATA folder	● From the provided source repo, download the "averages_year_month.csv" file and store in an easily located folder ● Filter the dataset to only include the rows that contain years between 2000-2020 and save as a new .csv file. ● Store your newly created, filtered dataset in the DATA folder of your case study repo
CODE folder	● Download EITHER the "evaluation.ipynb" or "sarima_modeling.Rmd" and open in your chosen IDE. ○ Note that both do the same evaluation (finding the best-fitting model), however the .Rmd uses methods that are less time-consuming compared to the .ipynb ● Change the dataset to your filtered .csv file and run in its entirety. ● Save the script (with the edits that you made) in the CODE folder of your case study repo
OUTPUT folder	● Include all output produced as a result of the analysis ○ Can be any format (pdf, jpeg, etc) ● In particular, any output used in your written report should be included in this folder

Acknowledgements: This structure is pulled from [Streifer & Palmer \(2020\)](#).